Week 1:

Introduction :=

In the first week of my leaning journey in WiDS and DS via the Spotify song mood classifier project, I have gained foundational knowledge in several key areas of data science. This includes probability and statistics, basic Python programming, using popular Python libraries like Numpy, Pandas, data visualization with Matplotlib, and introduction to ML.

1. Probability and Statistics

Probability and statistics are the building blocks for data analysis. Probability helps in understanding the likelihood of events, that how often can this event accrue, while statistics helps summarizing, interpreting, and analysing the data.

The concepts I learned in Prob. & Stats. :

- Probability: The chance of an event happening.
- Random Variables: Variables whose values depend on chance.
- Mean, Median, Mode: Measures of central tendency used to summarize data.
- Variance and Standard Deviation: Measures of how much the data spreads out from the mean.
- Distributions: Patterns that data follow, such as normal distribution, which is common in natural data.
- Correlation: Shows how two variables are related. A positive correlation means when one variable increases, the other also increases.

This thing helped me in understanding the “data patterns” and “trends in the data”.

2. Basics of Python

Python is a versatile programming language widely used in data science due to its simplicity and large ecosystem of libraries. Honestly saying it was very easy to learn and I was able to understand it very easily.

Topics that I’ve learned in the Python:

- Variables and Data Types: Stores information like numbers, text, and lists.
- Control Structures: Using if-else statements and loops (for and while) to control the flow of a program.
- Functions: used to perform specific tasks.

- Libraries: Python’s packages like Numpy, Pandas, and Matplotlib extend its capabilities for data science.

Python, beginner friendly and easy to learn language, helped me to apply Machine Learning on the data neatly and accurately in next weeks.

3. Numpy Library

Numpy is a Python library for numerical computing. It is fast and efficient for handling large arrays and matrices of numbers.

What I Learned in NumPy:

- Arrays: Numpy arrays are like lists but faster and more memory-efficient.
- Array Operations: We can do addition, subtraction, multiplication, and division on entire arrays at once.
- Mathematical Functions: Functions like sum, mean, and std help analyse data quickly.

Numpy is essential because many data science and machine learning libraries use it internally for calculations. NumPy makes calculations easy for the program.

4. Pandas Library

Pandas is a library for data analysis. It allows us to handle structured data efficiently.

What I learned while working with Pandas:

- DataFrames: 2-dimensional data structure similar to a table in Excel.
- Series: 1-dimensional array with labels.
- Reading Data: Importing data from CSV, Excel, or other formats using pd.read_csv and other codes.
- Data Cleaning: Handling missing data, filtering rows, and selecting columns.
- Data Operations: Grouping, sorting, and summarizing data to find trends.

Pandas makes it easier to prepare data for analysis and machine learning models.

5. Data Visualization with Matplotlib

Visualization is crucial to understand data patterns, trends and communicate insights effectively. Matplotlib is a popular library for creating graphs and charts in Python.

Another library that I used alongside Matplotlib is seaborn, which also had similar type things. It is also used to create graphical representations.

Topics that I covered in Matplotlib and other visual representation library:

- Plots and Graphs: Line charts, bar charts, and scatter plots to represent data.
- Customizing Graphs: Adding titles, labels, legends, and colours to make graphs more good looking and easy to read.
- Understanding Patterns: Visualization helps identify trends, outliers, and relationships between variables.

Visualization is a key step before applying machine learning models because it helps understand the data and spot problems. Graphical representations which are done by MatPlotLib and also SNS, help us in understanding even those data trends which might have not discovered / spotted of by statistics or probability. Like we can spot outliers by using the box plot.

6. Introduction to Machine Learning

Machine learning allows computers to learn from data and make predictions without explicit programming.

In the first week I learned some basic concepts of ML.

Machine learning builds on the knowledge of statistics and programming, making it a powerful tool for real-world applications like finance, healthcare, and marketing data.

Conclusion :=

In the first week, I learned the essential building blocks of data science. Understanding probability and statistics helped me analyze and interpret data, while Python and its libraries make data handling and computation easier. Data visualization allows us to communicate insights effectively, and an introduction to machine learning shows how data can be used to make predictions. These skills will serve as a foundation for deeper learning in the coming weeks.

Other than this I was given assignment based on what I've learn in the whole week. The assignment questions tested my understanding and skills regarding the topics that I've learned.

Week 2:

Introduction:=

In the second week of my learning journey, I explored more advanced topics in data science, focusing on Exploratory Data Analysis (EDA), data visualization, data cleaning and preprocessing, and the fundamentals of supervised machine learning. I also worked with real-world data, specifically the official Spotify audio features dataset, which helped me understand the characteristics of music tracks and their audio properties.

These skills are essential because real-world data is often messy, and extracting meaningful insights requires proper analysis, visualization, and preparation before applying machine learning models.

1. Exploratory Data Analysis (EDA)

Exploratory Data Analysis is the process of analyzing data sets to summarize their main characteristics, often using visualization and statistical methods. EDA is a crucial step because it helps us understand the data before modelling. EDA is very very useful and important when we want to create a model which is highly accurate.

Topics that I've learned in EDA:

- Introduction to EDA: Understanding the structure of the dataset.
- Identifying missing values, duplicates, and incorrect data types.
- Getting descriptive statistics like mean, median, standard deviation, and distribution of data.
- Complete EDA Workflow in Python: Using Pandas to explore data (head(), info(), describe() methods).
- Checking for null values and basic data cleaning.
- Understanding relationships between variables using correlation matrices.
- Visualization Tools: Matplotlib Basics: Creating line charts, bar charts, and scatter plots to visualize trends and distributions.
- Seaborn for Statistical Plots: Creating advanced plots like boxplots and heatmaps to understand distributions and relationships between features.

2. Understanding Spotify Audio Features

Working with Spotify's audio features dataset introduced me to music analytics and the characteristics of sound in data science. I been through the official datasheet of the songs which are available on the Spotify application. Although the data happens to be of 2019, it contained a very large number of songs in it.

The spotify Audio Features that I Learned:

- Danceability: How suitable a track is for dancing.
- Energy: Measures intensity and activity of the track.
- Loudness: Overall volume in decibels.

- Acousticness: Probability that the track is acoustic.
- Valence: Musical positiveness or happiness in a track.
- Tempo: Speed of the track in beats per minute.

More detailed information about these and many other features were available on the site which clearly explained role of such feature in a song and also song's mood's dependency on it.

This helped me analyze musical trends, recommend songs, or even predict popularity based on audio characteristics. Working with this dataset also helped me practice EDA and data visualization on real-world data.

3. Data Cleaning and Processing

Data cleaning and preprocessing are critical because raw data is often incomplete, inconsistent, with visible duplicity of some data or in a format that machine learning models cannot understand.

What topic was there in Data cleaning and processing:

- Handling Missing Values: Identifying missing data in datasets.
- Filling missing values with mean, median, or mode.
- Removing rows or columns with too many missing values.
- Feature Scaling: Ensuring features are on the same scale to improve model performance.
- Normalization: Scaling data to a range of 0 to 1. (This was a specific needed thing in spotify dataset as it contained many such features whose range was very large as compared to other available features.)
- Standardization: Transforming data so it has a mean of 0 and a standard deviation of 1.
- Data Transformation: Converting categorical variables into numerical forms using encoding techniques. It was a good to learn topic but somehow I struggled learning it.

These steps makes sure that data is ready for machine learning models and help improve accuracy and performance.

4. Introduction to Supervised Machine Learning

Supervised learning involves teaching a model using labelled data, where the input features and the corresponding output labels are known. The model learns patterns to predict future outcomes.

Algorithms that I Learned briefly:

- Linear Regression: Predicts a continuous target variable using one or more input features.
- Logistic Regression: Predicts binary outcomes, such as whether a song is popular or not.
- K-Nearest Neighbors (KNN): Predicts the target value by looking at the nearest data points.
- Simple and intuitive for classification tasks.
- Decision Trees: Uses a tree structure to make decisions based on input features.
- Easy to interpret and visualize.
- Different types of Model Evaluation that I’ve looked in:
- Train-Test Split: Dividing data into training and testing sets to evaluate model performance. Mostly in my assignments I’ve used Train-Test Split in 80-20 ratio.
- Confusion Matrix: Shows the number of correct and incorrect predictions in classification problems.
- Performance Metrics: Accuracy, precision, recall, and F1-score help evaluate how well the model performs.

I used Train-Test split and Cunfusion matrix many times in my over all WiDS project and it helped me understand the model more better. Although it took me a bit longer to Train the model as data was large and process was lengthy.

5. Summary of Practical Work

During this week, I applied these concepts to the Spotify dataset. My workflow included:

- Loading Data: Using Pandas to read CSV files and examine the structure.
- Performing EDA: Visualizing distributions and relationships with Matplotlib and Seaborn.
- Cleaning Data: Handling missing values and scaling features.
- Building Models: Implementing linear regression, logistic regression, KNN, and decision trees.
- Evaluating Models: Using train-test split, confusion matrix, and metrics to measure performance.

This practical experience helped me connect theory with real-world data analysis and machine learning.

Conclusion:=

Week 2 was highly productive and strengthened my foundation in data science. I learned how to explore and visualize data, understand audio features of Spotify tracks, clean and preprocess data, and apply supervised machine learning algorithms.

These skills are critical because they prepare data for analysis, allow us to identify patterns, and make accurate predictions. The hands-on practice with real-world datasets like Spotify makes learning more engaging and practical, setting a strong base for more advanced topics in data science and machine learning.

I was given an assignment at the end of week 2 also. It tested my ability to work with real world data and ability of dealing with the data which requires data cleaning and data scaling. It helped me realise that real world data is very diverse and can contain many errors and mistakes.

Week 3:

Introduction :=

In Week 3 of my data science learning journey, I focused on unsupervised learning, which is a type of machine learning where the model finds patterns in data without labeled outputs. I also learned about clustering techniques, including hierarchical clustering, k-means clustering, and Gaussian Mixture Models (GMM).

Additionally, I explored advanced concepts like feature selection, feature engineering, hyperparameter tuning using Grid Search CV, and model evaluation metrics later. These topics are important because they help improve model performance and make data analysis more efficient. The topics which was covered in this week and also after that played a very important part in my final model.

1. Unsupervised Learning

Unsupervised learning is used when we do not have labelled data. The goal is to find patterns, trends, similarities, or groups in the data.

The topic I learned in unsupervised learning :

- Clustering: Grouping similar data points together based on their features. Clustering is a complex process. At first, the data may seem to have only a few clusters, but a deeper look can reveal more clusters.

Unsupervised learning is useful when we want to find hidden patterns in data that are not obvious at first glance, and clustering is one of the main techniques used for this.

2. Hierarchical Clustering

Hierarchical clustering builds a tree of clusters called a dendrogram. It groups data points step by step, either from bottom-up (agglomerative) or top-down (divisive).

Key Concepts which included in Hierarchical Clustering are:

- Agglomerative Clustering: Starts with each point as its own cluster and merges the closest clusters iteratively.
- Divisive Clustering: Starts with all points in one cluster and splits them iteratively.
- Dendrograms: Visual representations showing how clusters are merged or split at each step.
- Clustering comes with many Advantages as it helps spotting groups and trends but it also have some disadvantages.
- Advantages are:
- Does not require specifying the number of clusters initially.
- Easy to visualize relationships between clusters.
- Disadvantages are:
- It can be slow for large datasets.
- Sensitive to noise and outliers.

The data that I used in spotify was happened to be very large that I was unable to perform hierarchical clustering on the whole data set at once. So, to perform the Hierarchical clustering, I have to take little sample of the data.

3. K-Means Clustering

K-means clustering is a widely used partitioning method where data is divided into a fixed number of clusters (K). The main advantage of this clustering method is that it can be performed on even very very large dataset.

Algorithm Steps of K-Means:

1. Choose K initial cluster centers.
2. Assign each data point to the nearest centroid.
3. Recalculate centroids based on assigned points.
4. Repeat steps 2–3 until centroids do not change significantly.

Key Points Learned:

- Choosing the correct number of clusters is critical. Techniques like the Elbow Method help determine K. We can choose an optimal K after performing the Elbow method.
- K-means is efficient for large datasets but assumes clusters are spherical and of similar size which might not 100% true. Clusters can be uneven instead of being fully spherical.

I used the K-Means clustering for grouping similar songs in Spotify's dataset based on audio features like energy, tempo, and danceability.

4. Gaussian Mixture Models (GMM)

Gaussian Mixture Models are a probabilistic clustering method. Instead of assigning each point to a single cluster, GMM calculates the probability of belonging to each cluster.

What is the concept of GMM:

- Each cluster is represented as a Gaussian (bell curve) in multiple dimensions.
- Points can belong to multiple clusters with different probabilities.
- It is so useful when clusters have overlapping boundaries. This is the exact problem that we were facing in the spotify dataset as i've grouped the songs in different moods but moods like happy and very happy happens to have very thin line of boundry.
- Advantages of GMM:
- Handles non-spherical clusters better than K-means.
- Provides probabilities, giving more information about cluster membership.

About learning extra resources:=

Along with clustering, I explored additional resources to strengthen my data science skills and build a good working model based on the provided Spotify dataset. These are the topics which were neccesory for developing the model but because of the time boundry, I was not able to explore it at it's fullest.

1. Feature Selection:

- Choosing the most important features to improve model performance and reduce overfitting.
- Techniques include correlation analysis and statistical tests.

2. Feature Engineering Basics:

- Creating new features or modifying existing ones to better represent data.
- Examples: Combining multiple audio features to create a "popularity score" for songs.

3. Grid Search CV (Cross-Validation):

- Technique to find the best hyperparameters for a machine learning model.
- Uses cross-validation to evaluate performance for each combination of parameters.

4. Model Evaluation Metrics:

- Understanding how well a model performs is crucial.
- Metrics for classification: accuracy, precision, recall, F1-score.
- Metrics for regression: mean squared error (MSE), mean absolute error (MAE), R^2 score.

These topics are important for optimizing models and making better predictions.

Summary of Practical work:=

=During Week 3, my workflow included:

- Exploring Un-labeled Data: Using Python and Pandas to understand data patterns.
- Applying Clustering: Hierarchical clustering for small datasets and dendrogram visualization of it.
- K-means clustering for dividing data into K clusters.
- Gaussian Mixture Models to handle overlapping clusters.
- Learning Model Improvement Techniques: Practicing feature selection and feature engineering.
- Exploring Grid Search CV for hyperparameter tuning.
- Understanding Evaluation Metrics: Using metrics to compare models and ensure their reliability.

I was assigned an assignment at the end of week 3 also and it was basically to test if I understands concepts such as clustering and Model evaluation. This part of the project was very crucial as it contained main resources to create a strong and accurate model in a real world. As a data scientist it was a must do part and, yes questions designed in the assignment was of that level that it tested my ability to work on such model and clustering.

Conclusion:=

Week 3 expanded my knowledge in unsupervised learning and clustering, helping me analyze data without labels. I learned hierarchical clustering, K-means clustering, and Gaussian Mixture Models, each with its advantages and use cases.

The last week, additionally, exploring feature selection, feature engineering, Grid Search CV, and model evaluation metrics gave me insight into how data scientists improve models and make reliable predictions.

These skills are essential for discovering hidden patterns in data and for preparing myself for more advanced topics in machine.

At the end of my 3 weeks learning journey, I have gained a very good knowledge of Data science and data analysis and also at the end of everything I was able to create such a model that all the topics that I've learned so far can be applied on. The assignment during each week was already helping me understand better and at last the final project gave good amount of confidence regarding my knowledge.